AI.112

Computer Graphics

COURSE OVERVIEW

Generating synthetic images: geometry, transformations, lighting, rendering, and visualization for simulation.



ILLUSTRATIVE HERO IMAGERY

COMPUTER GRAPHICS

1

SESSION

60 MINUTES

ORIENTATION: PIXELS FROM MATH

KEY TOPICS

- Raster vs. vector graphics
- The graphics pipeline
- Coordinate systems
- Graphics in simulation and VR

TEACHING FOCUS

Create images algorithmically rather than capturing them.

KEY TAKEAWAY Graphics synthesize reality from geometry.

2

SESSION

60 MINUTES

2D & 3D TRANSFORMATIONS

KEY TOPICS

- Translation, rotation, scaling
- Homogeneous coordinates
- Composite transformations
- Viewing and projection

TEACHING FOCUS

Move and shape objects in virtual space.

KEY TAKEAWAY Transformations place objects in a scene.

3

SESSION

60 MINUTES

GEOMETRIC MODELING

KEY TOPICS

- Polygon meshes
- Curves and surfaces
- Scene graphs
- Modeling for robot simulation

TEACHING FOCUS

Represent shape digitally.

KEY TAKEAWAY Meshes are the skin of virtual objects.

4

SESSION

60 MINUTES

LIGHTING & SHADING

KEY TOPICS

- Light sources and materials
- Phong and Gouraud shading
- Shadows
- Global illumination intro

TEACHING FOCUS

Make virtual scenes look physical.

KEY TAKEAWAY Lighting gives form its presence.

5

SESSION

60 MINUTES

RENDERING & VISUALIZATION

KEY TOPICS

- Rasterization and ray tracing basics
- Textures and mapping
- Real-time rendering
- Visualizing sensor data

TEACHING FOCUS

Produce the final image from a scene.

KEY TAKEAWAY Rendering is the last step of the pipeline.

6

SESSION

60 MINUTES

GRAPHICS FOR ROBOTICS & REVIEW

KEY TOPICS

- Robot simulation environments
- Visualizing point clouds
- Course recap and assessment
- Toward VR (AI.301)

TEACHING FOCUS

Use graphics to simulate and test robots.

KEY TAKEAWAY Simulation lets robots practice before they move.